

Qatar-1b

R-0632

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-1_230622 · created 2026-07-25T16:27:05+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2460117.8688 +/- 0.0025 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.192 +/- 0.04
Transit depth (Rp/Rs)^2	3.7 +/- 1.53 [%]
Orbital Inclination (inc)	84.47 +/- 2.01
Airmass coefficient 1 (a1)	0.842 +/- 0.079
Airmass coefficient 2 (a2)	0.15 +/- 0.27
Scatter in the residuals of the lightcurve fit is	9.38 %
Best Comparison Star	#1 - [627, 437]
Optimal Aperture	3.57
Optimal Annulus	17.12
Transit Duration (day)	0.072 +/- 0.013

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.192 $\pm$ 0.04 vs 0.1463 (deviation 0.9 σ)
Duration fit vs published	0.072 d vs 0.0692 d (deviation 0.17 σ)
Mid-time O-C	-1.7 min
Depth SNR	3.8
Residual scatter	9.38 %

Light curve

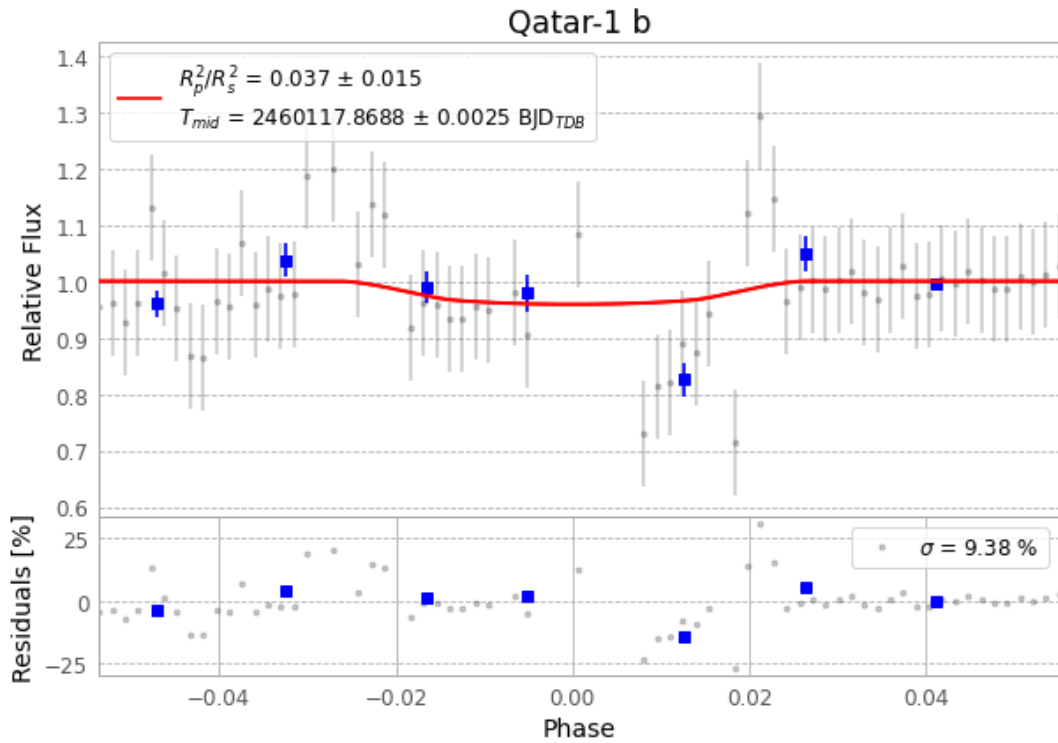


Figure 1 — FinalLightCurve_Qatar-1 b_2023-06-22.png (SHA-256 70cb143e06b36137...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [291, 218], 3 comparison star(s). Exact configuration: parameters.inits.json in record.json (SHA-256 6352c356bfec8125...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit b365494

Data provenance

75 FITS frames (49 MB), Qatar-1230622070015.FITS → Qatar-1230622104215.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-1-230622.log (d0b94b4a332a...), FinalLightCurve_Qatar-1 b_2023-06-22.png (70cb143e06b3...), FinalLightCurve_Qatar-1 b_2023-06-22.csv (caf3f8208065...)

Compute resources

Wall time 190.4 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-25 16:27Z.